Design and Methodology Report

EV Battery Health Prediction Using Limited and Noisy Sensor Data

1. Overview

This project evaluates how the accuracy of machine-learning (ML) battery-health models changes as sensor data quality decreases. Public battery-ageing data is used as a clean baseline, controlled degradation is applied, and the resulting prediction error is measured for **State of Health (SOH)** and **Remaining Useful Life (RUL)**.

Objective: determine the minimum data quality required for reliable battery-health prediction.

2. System Design

The system is a data-quality stress-testing pipeline with the following stages:

Stage	Description
1. Data acquisition	Load public battery-ageing datasets
2. Preprocessing	Clean data and compute SOH and RUL labels
3. Cycle segmentation	Split records into individual charge/discharge cycles
4. Data-quality simulator	Apply controlled degradation to the data
5. Feature extraction	Derive health-related features from each cycle
6. Model training	Train baseline and ML models
7. Prediction	Estimate SOH and RUL on unseen batteries
8. Evaluation	Compute error metrics
9. Robustness analysis	Relate prediction error to degradation level

3. Datasets

Role	Dataset	Purpose
Primary	NASA PCoE Battery Dataset	Model development and main experiments
Secondary	CALCE CS2	Check whether findings hold on a different dataset

The data includes voltage, current, temperature, time, capacity, and charge/discharge cycle records.

4. Data-Quality Simulator

The simulator is the central contribution of the project. Rather than using only clean laboratory data, it deliberately degrades measurements to imitate real vehicle sensor conditions.

Degradation factor	Simulated condition	Real-world cause
Reduced sampling rate	Fewer readings per cycle	Slower BMS data logging
Sensor noise and rounding	Random error added to voltage, current, and temperature	Lower-precision or lower-cost sensors
Missing data	Readings randomly removed	Dropped measurements, communication gaps
Incomplete charging	Only part of the charging curve retained	Partial or interrupted charging sessions

Each factor is applied at several severity levels (for example, noise standard deviation, fraction of readings removed, sampling interval, or portion of the charge curve retained).

5. Feature Extraction

Features are derived from each cycle before modelling:

- Voltage statistics
- Charging time
- Maximum temperature
- Temperature rise
- Charge-based capacity
- Cycle information

These capture the degradation patterns that the models learn from.

6. Machine-Learning Models

Model	Role
Cycle-count-only predictor	Naive baseline: tests whether ML adds value beyond simply knowing the cycle number
Ridge Regression	Simple, interpretable linear baseline
Random Forest	Captures nonlinear relationships between features and health
XGBoost / Gradient Boosting	Stronger tree-based model

7. Prediction Targets

State of Health (SOH) describes the battery's current condition:

> *SOH (%) = (Current capacity / Initial capacity) × 100*

Example: initial capacity 2.0 Ah, current capacity 1.7 Ah, so SOH = 85%.

Remaining Useful Life (RUL) is the number of cycles remaining until SOH falls to 80%, the end-of-life threshold used in this project.

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8. Validation Strategy

Models are evaluated with **leave-one-battery-out validation**. Each battery in turn is held out as the test set while the remaining batteries form the training set:

Fold	Training	Testing
1	Batteries 1, 2, 3	Battery 4
2	Batteries 1, 2, 4	Battery 3
3	Batteries 1, 3, 4	Battery 2
4	Batteries 2, 3, 4	Battery 1

Randomly splitting cycles from the same battery across training and testing sets can substantially overstate performance. Holding out entire batteries gives a realistic estimate of how a model performs on a battery it has never seen.

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9. Performance Metrics

Metric	Meaning	Use
MAE (Mean Absolute Error)	Average size of prediction error	SOH (%) and RUL (cycles)
RMSE (Root Mean Square Error)	Average error, penalising large errors more heavily	SOH (%) and RUL (cycles)

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10. Robustness Analysis

This is the main output of the project. Each degradation factor is increased in steps from clean data to severe degradation, and prediction error is recorded at every level.

Factor	Example progression
Noise	Clean → low → medium → high
Missing data	0% → low → medium → high fraction removed
Sampling rate	Full → reduced → sparse
Charging completeness	Full curve → partial → short segment

Results are plotted as prediction error against degradation level for each model and factor. The point where error exceeds a predefined tolerance marks the **minimum data quality required for reliable prediction**.

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11. Complete Methodology

1. Collect public battery data.
2. Clean and preprocess the data.
3. Calculate SOH and RUL.
4. Extract battery features.
5. Create degraded datasets at multiple quality levels.
6. Train the ML models and the baseline.
7. Test on unseen batteries.
8. Calculate MAE and RMSE.
9. Compare model robustness.
10. Determine the minimum required data quality.

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12. Expected Outcomes

The study is designed to answer four questions:

1. Which ML model performs best?
2. Which features remain informative as data quality decreases?
3. How quickly does prediction accuracy deteriorate?
4. What is the minimum data quality needed for reliable prediction?

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13. Project Summary

> *We take clean battery-ageing data and deliberately make it noisy, incomplete, and less frequently sampled. We then train several machine-learning models to predict SOH and RUL, and measure how their performance changes. Our goal is to find the minimum data quality needed for reliable battery-health prediction.*